



POLITÉCNICA

Quantum-inspired Technologies For Smart Grids

BY: KYOUNG-TAE (ELIJAH), KIM

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The Smart Grid Challenge Landscape

The Smart Grid Challenge

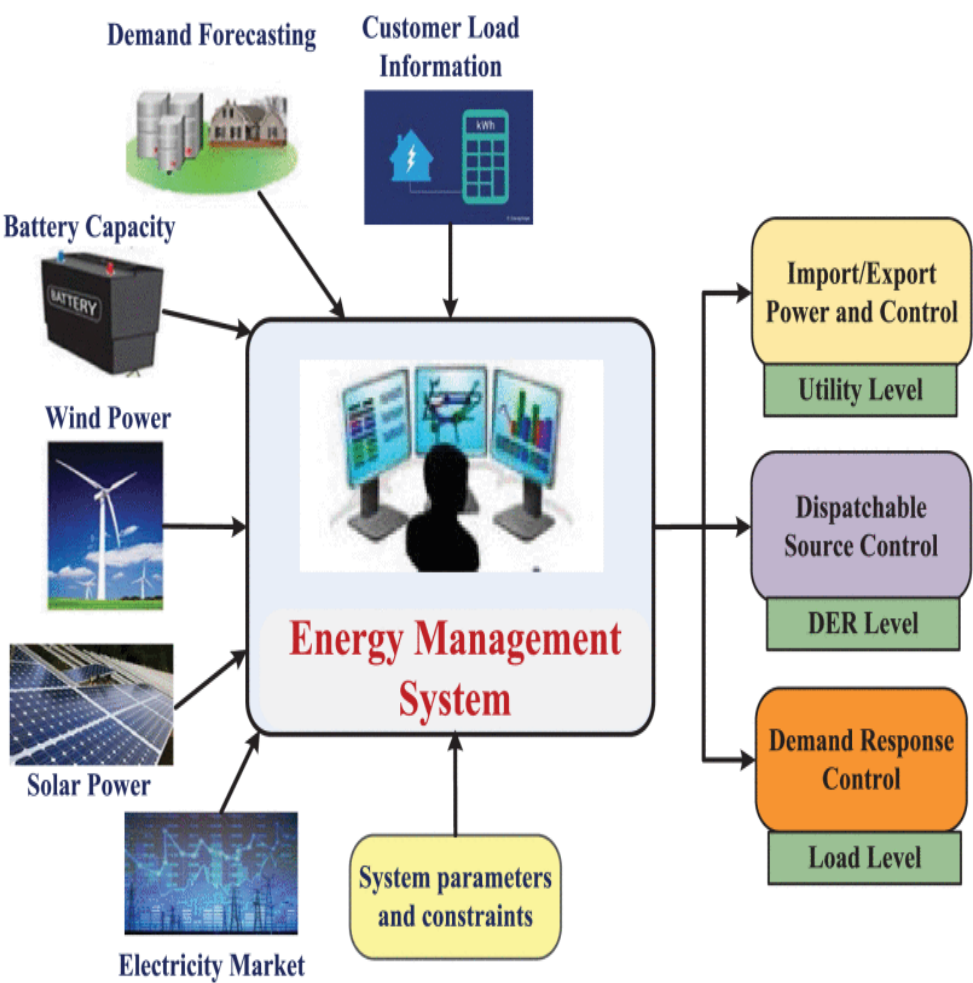
- **Renewable Integration:** 45% DER penetration by 2025 creates volatility
- **Cybersecurity Threats:** 65% increase in grid cyberattacks since 2020
- **Quantum Threats:** Shor's algorithm breaks current encryption
- **Computational Bottlenecks:** Real-time optimization exceeds classical limits

Quantum-Inspired Motivation

- Enhanced modelling accuracy for grid dynamics
- Quantum-resistant cryptographic security
- Exponential speedup for complex optimization

Energy Management with Quantum Optimization

Classical Limitation	Quantum-Inspired Solutions
Unit Commitment with 1000+ generators: 6-8 hours computation time.	Quantum Annealing for Optimal Power Flow: Achieves up to 3x faster convergence for real-time dispatch.
Real-Time Pricing with DERs: Often results in suboptimal, inefficient solutions.	Quantum TLBO for Microgrids: Demonstrated 15% cost reduction in dynamic pricing and resource allocation.
EV Charging Coordination: Faces severe scalability issues with growing EV adoption.	Quantum Reinforcement Learning for E-Mobility: Achieved 89% utility maximization in charging coordination.
Prosumer Problem : manage the generation, storage, consumption, and sharing of energy within a community?	Quantum Approximate Optimization Algorithm (QAOA), significant speedups, approach is beneficial for large-scale energy communities, providing high-quality solutions in reduced timeframes.

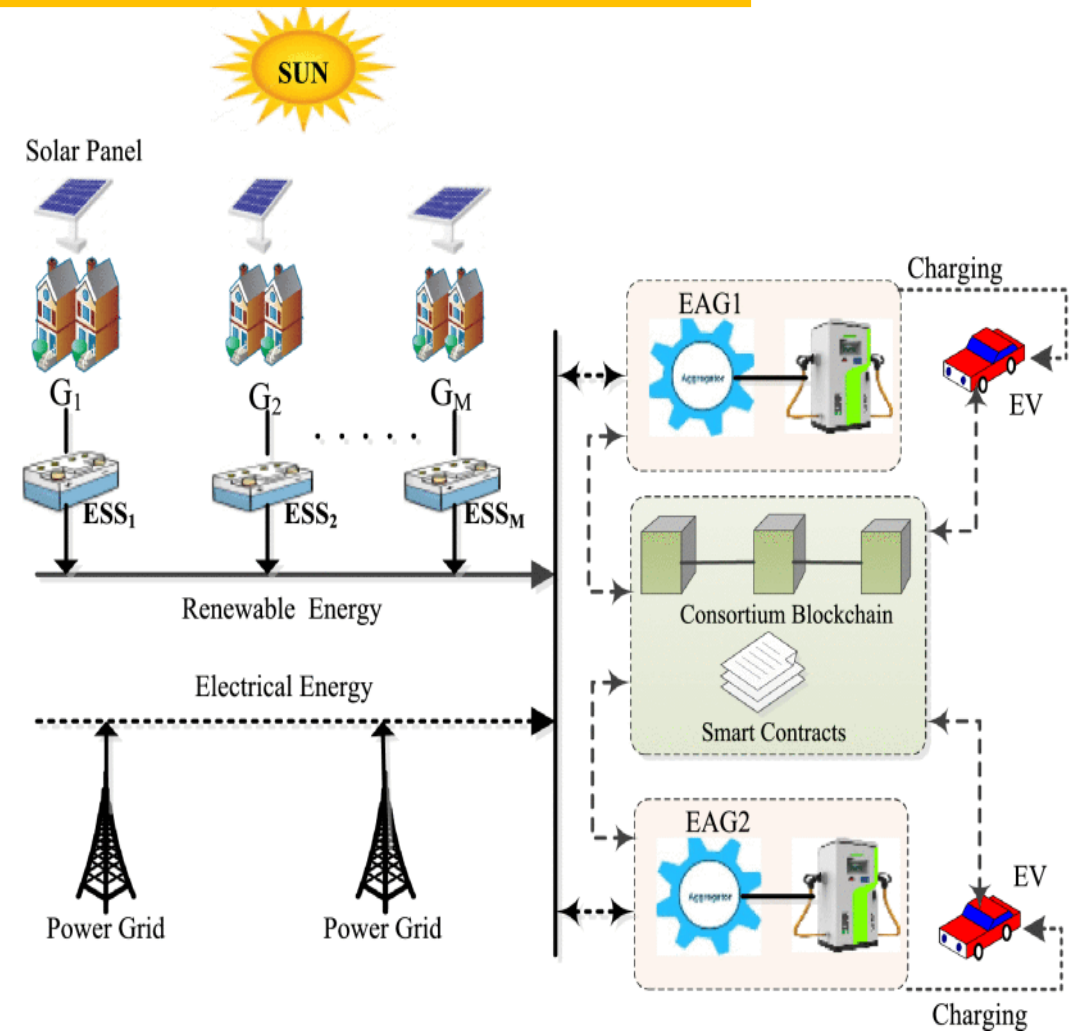


Quantum-Enhanced Energy Trading & Markets

Trading Platform Architecture:

Blockchain + Quantum Integration:

- **Smart Contracts:** Automated, transparent trading
- **Quantum Reinforcement Learning:** Adaptive pricing strategies
- **Quantum-Resistant Crypto:** Secure transaction records



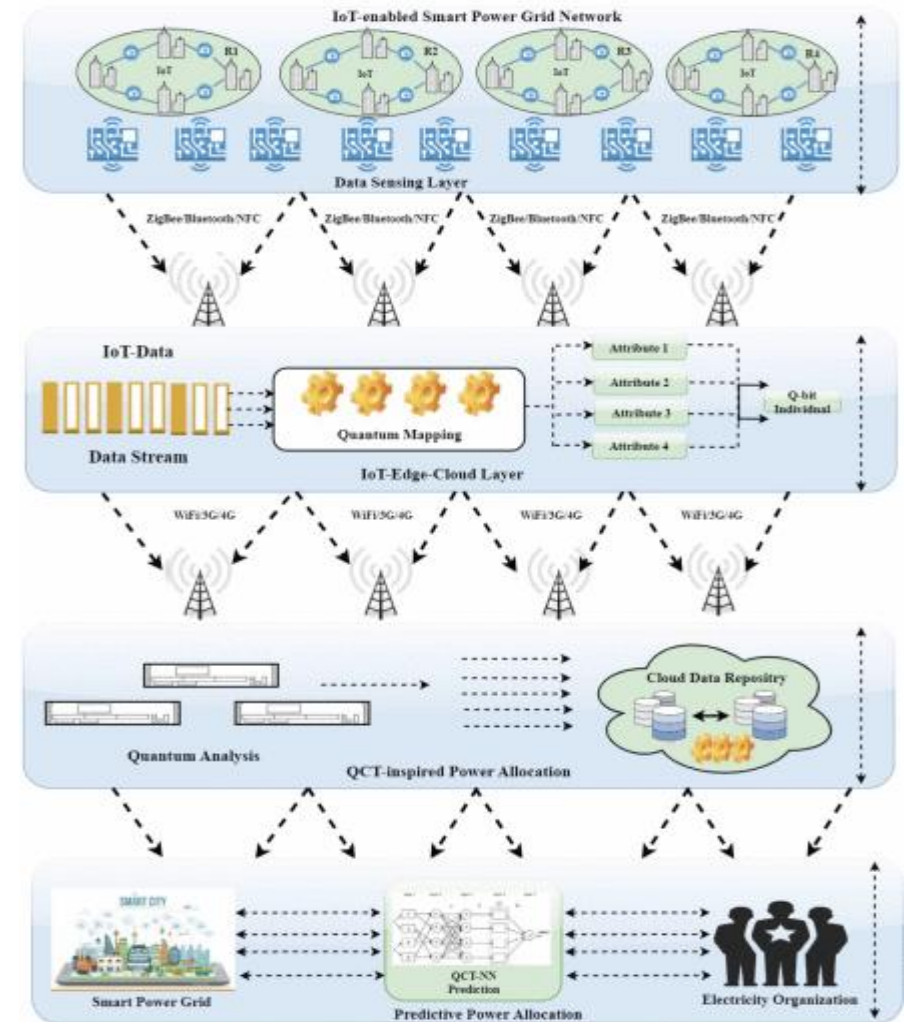
Edge Computing with Quantum-Inspired Intelligence

Quantum based Edge Framework:

- **Spatial-Temporal Utilization Index (STUI):** Real-time load forecasting
- **Automated Quantum-inspired Neural Network (AQNN):** Predictive allocation
- **Edge Deployment:** Reduced latency from 15.2ms to 5.9ms

Real-World Impact:

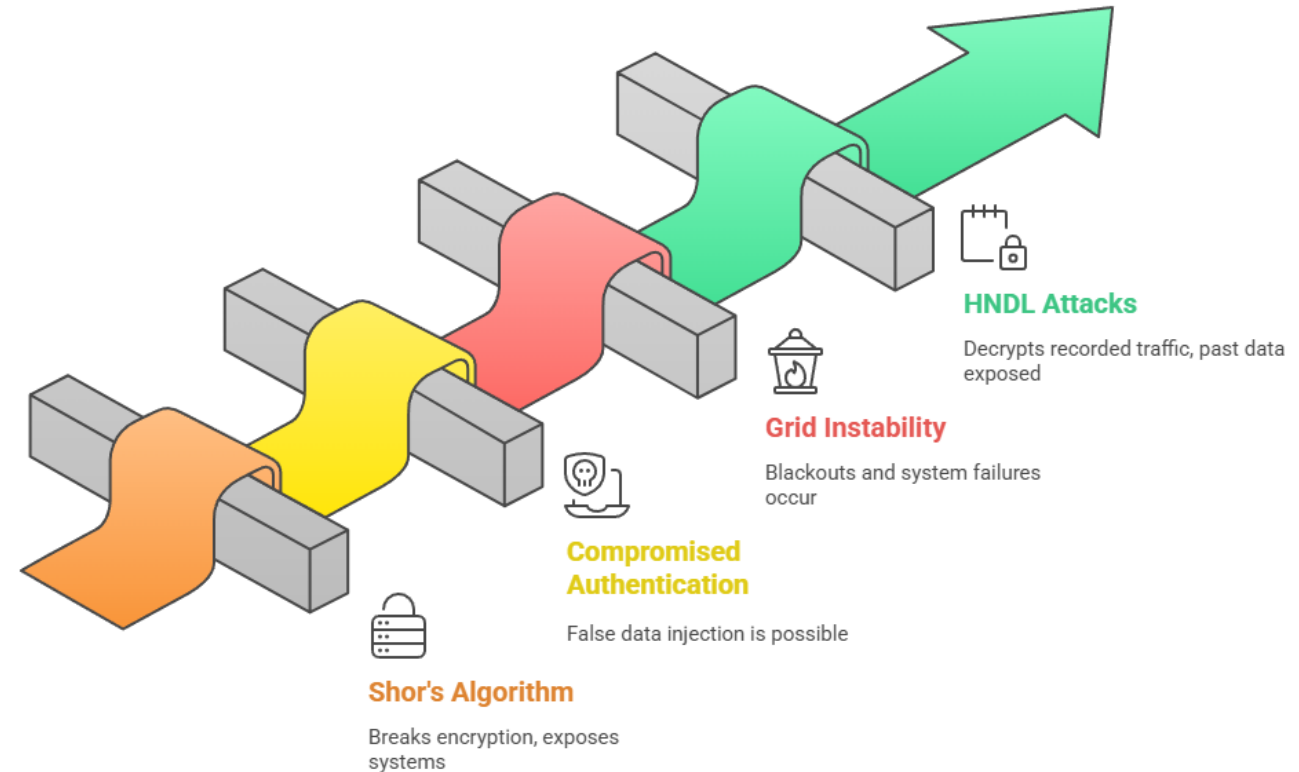
- **Smart Homes:** 25% better load balancing
- **Grid Substations:** 30% improved resource utilization
- **Renewable Integration:** 40% more accurate forecasting



Smart Grid Privacy & Security

Why This Matters

- Harvest Now, Decrypt Later" (HNDL) Risk
 - ✓ Grid communications often have **long secrecy lifetimes**. Grid topology data is **sensitive for decades**.
 - ✓ Hackers can record encrypted traffic today and decrypt it later.
 - ✓ This makes current smart grid deployments retroactively vulnerable.
- Long asset lifespans & Legacy Integration (RSA-1024)



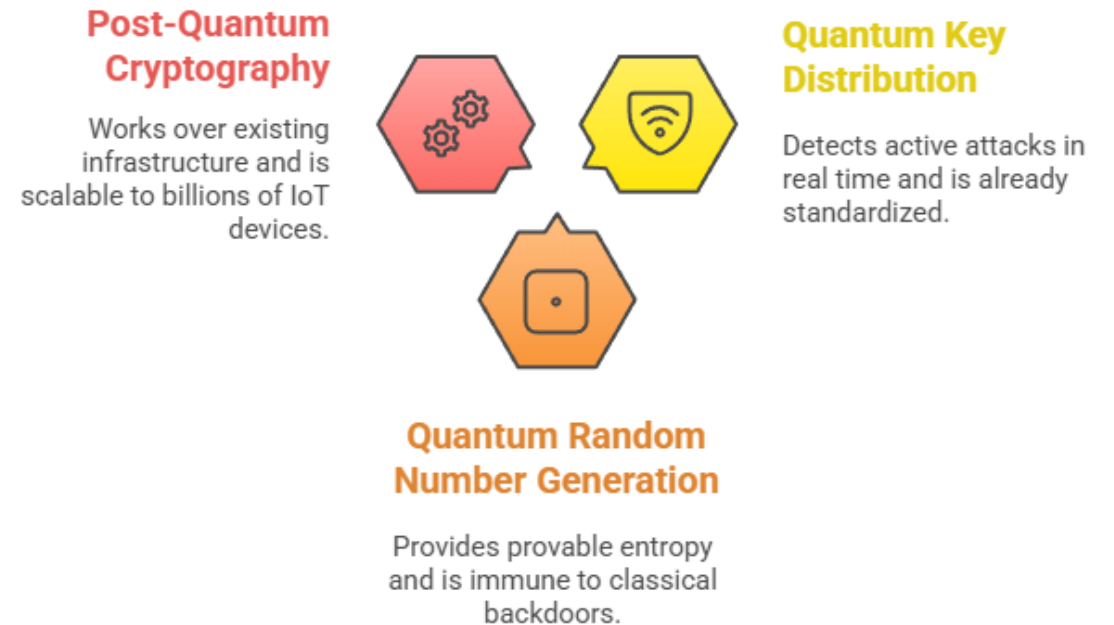
Shor's algorithm can break **RSA**, **ECSDA/ECDH** (meter authentication), and **DSA** (firmware signing) protocols

Smart Grid Privacy & Security

Three Quantum Security Approaches

- **Quantum Key Distribution (QKD)**
 - ✓ Uses **quantum states** to distribute a symmetric key with information-theoretic security
- **Quantum Random Number Generation (QRNG)**
 - ✓ Extracts **true randomness** from quantum measurement (e.g., photon arrival time, vacuum fluctuations).
- **Post-Quantum Cryptography (PQC)**
 - ✓ Classical algorithms **resistant to Shor's & Grover's algorithms** (lattice, hash, code, multivariate, isogeny).

Key Advantages



Smart Grid Privacy & Security

Visual Decision Matrix (Smart Grid Context)

Requirement	QKD	QRNG	PQC
Long-distance (100+ km)	✓ (with satellites/trusted nodes)	N/A	✓
Zero-trust key exchange	✓ (information-theoretic)	✗ (only randomness)	✗ (computational)
Retrofit existing meters	✗ (needs fiber)	✓ (chip-level)	✓ (firmware update)
Real-time control (<10 ms)	⚠ (latency ~ms-s)	✓	⚠ (PQC adds ~1-10 ms)
Cost per endpoint		\$	\$\$

Smart Grid Privacy & Security

Real-World Deployments (2025)

Technology	Example	Status
QKD	• China: 12,000 km Beijing–Shanghai QKD backbone (Micius satellite link). • Europe: EuroQCI (27-nation quantum network, 2027 target). • Toshiba/IDQ: Commercial QKD links in London, Geneva banks.	Production
QRNG	• ID Quantique: QRNG PCIe cards in HSBC, SK Telecom. • QuintessenceLabs: qStream 10 Gbps appliance.	Production
PQC	• Google: Hybrid Kyber in Chrome 124+ (2024). • AWS: PQ KMS with Kyber (2025). • Smart Grid Pilots: Southern California Edison testing ML-DSA in AMI (2024).	Standardization → Rollout

Smart Grid Privacy & Security

Limitations on Quantum Solutions

Solutions	Core Limitation	Smart Grid Impact
Quantum Key Distribution (QKD)	• Point-to-point only • ≤ 300 km fiber / 1,200 km satellite • No authentication • \$50K–\$500K per link • < 1 Mbps key rate	Unfeasible for 100M+ meters Only for core backbone
Quantum Random Number Generator (QRNG)	• Only entropy, not encryption/signing • Hardware trust vulnerable • Needs health tests	Seeds keys - not a full solution
Post-Quantum Cryptography (PQC)	• 10–100× larger keys/signatures • 10–100× slower • Legacy devices incompatible • New math = evolving risks	Only scalable fix Requires 5–10 yr migration

Smart Grid Privacy & Security

Quantum Security Applications

Qubitekk (United States)

- Demonstrated the world's first quantum key-based authentication system operating on a live utility fiber network.
- Provides QKD systems for SCADA and distributed energy resources (DER).
 - **United States:** EPB utility in Chattanooga, Tennessee - first commercial quantum network
 - America's first commercially available quantum network for smart grid (operational since 2016)



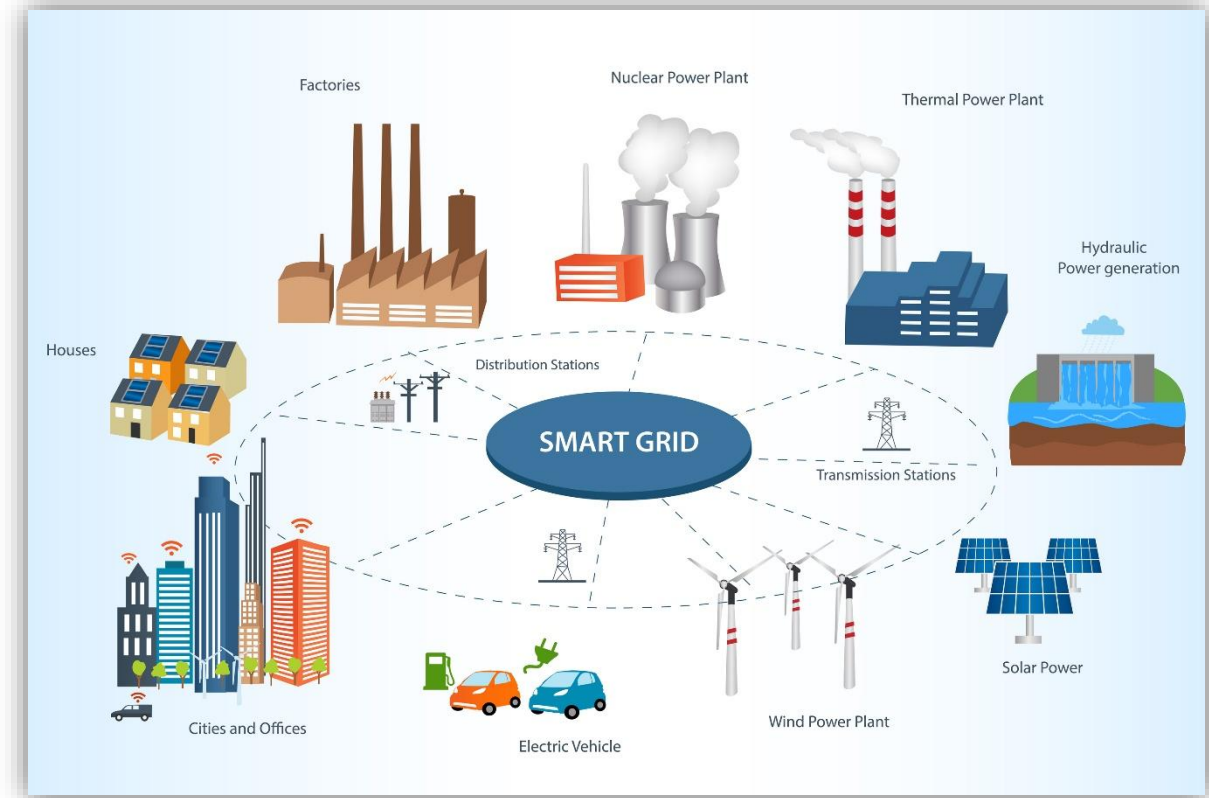
ID Quantique (Switzerland)

- Provides commercial quantum key distribution (QKD) systems specifically designed for smart grid communications infrastructure.
 - **Switzerland:** Geneva's power substations (since 2016) - securing 800+ power stations.
 - **South Korea:** 40km power communication network between substations in South Chungcheong Province with KEPCO.
 - **Austria:** Medical data transfer (demonstrating cross-sector capability)

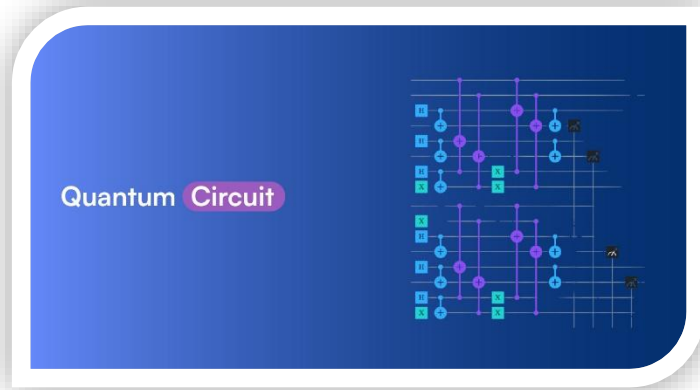


Power System Modelling: A Quantum Perspective

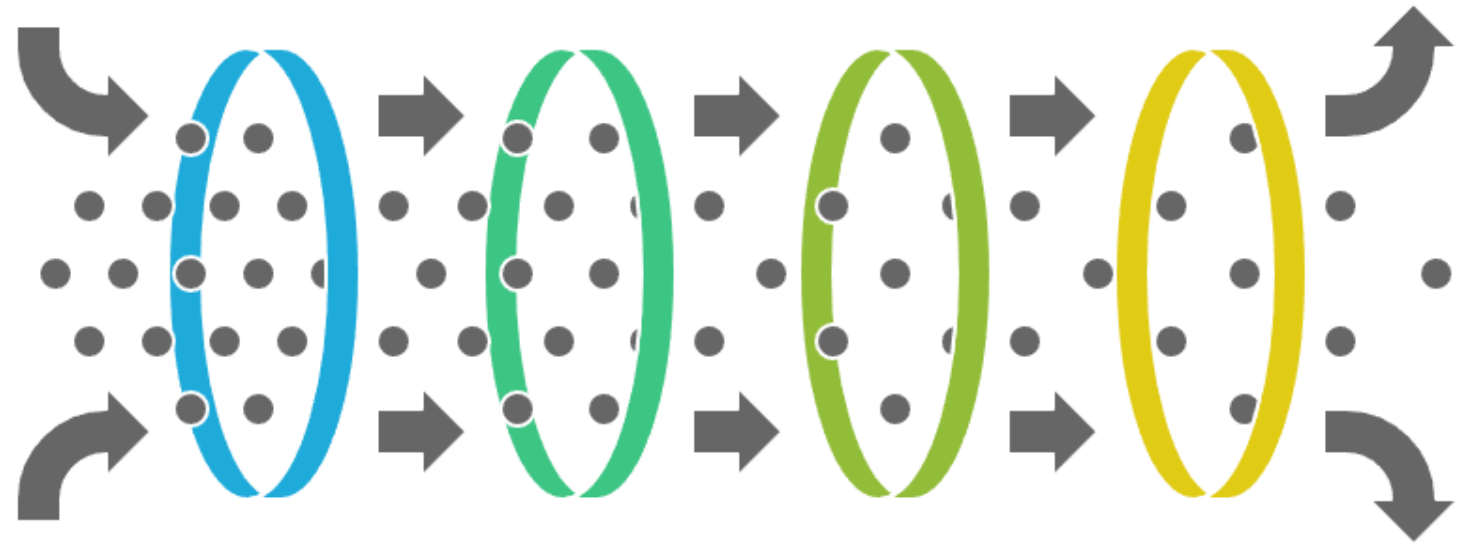
- **Modelling** = foundation for control and optimization.
- **Classical methods** face computational limits.
- **Quantum computing enables** new ways to simulate and predict grid behavior.



The HHL Algorithm for Predictive Control in Microgrids



HHL Algorithm Efficiency Funnel



Exponential Speedup

Achieves faster computation compared to classical methods

Model Predictive Control

Applies to control systems in microgrids

Accuracy Improvement

Enhances the precision of control models

Computational Efficiency

Reduces resource usage in computations

Hybrid and Distributed Quantum Approaches for Smart Grid Optimization

Quantum–Classical Co-Simulation for Smart Grids

- Proposed by Vereno et al. (2023).
- Combines classical and quantum simulation.
- Increases accuracy in energy system analysis.
- Supports better planning and optimization.

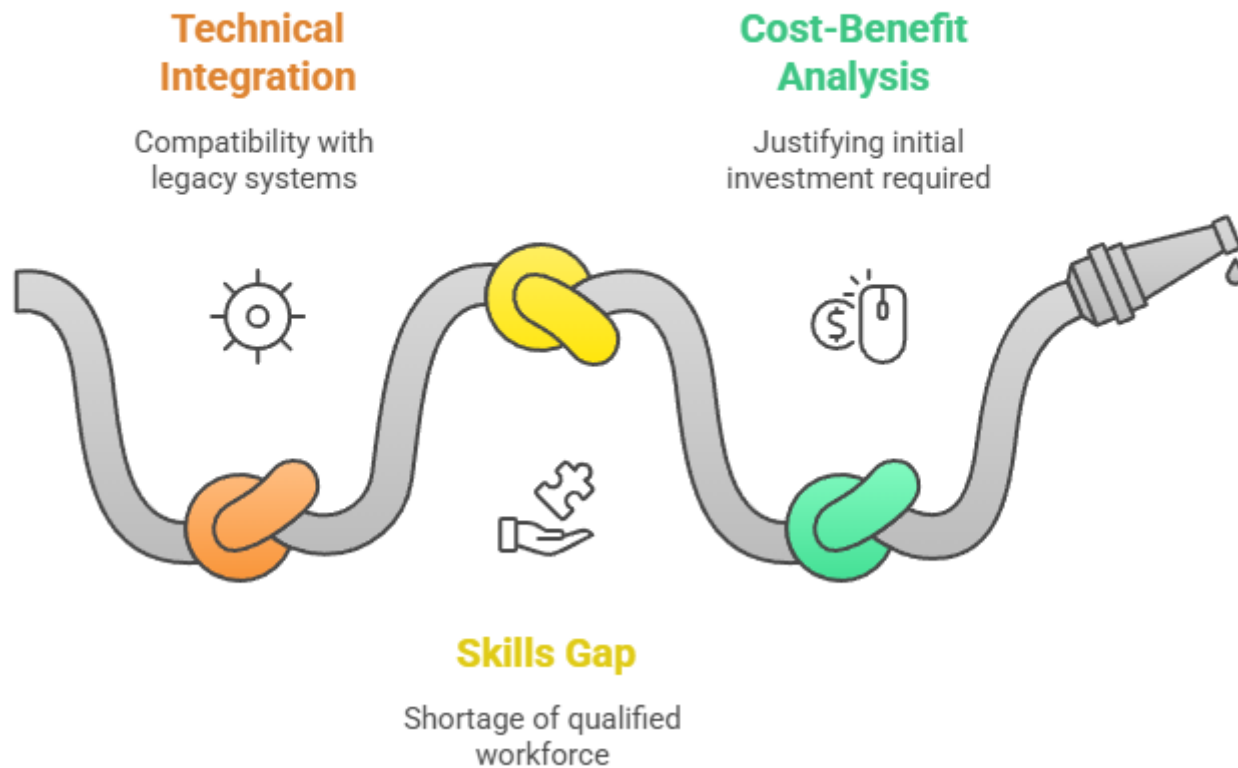


Collaborative Optimization through Distributed Quantum Computing

- Proposed by Ngoenriang et al. (2024).
- DQC2O: Distributed Quantum Computing for Collaborative Optimization.
- Distributes computation across multiple quantum processors.
- Enhances scalability and collaboration among energy sectors.

Challenges & Future Outlook

Quantum Grid Implementation Hurdles



Priority Research & Collaboration Opportunities:

- **Algorithm Enhancement:** Scaling quantum algorithms (e.g., HHL) for larger, more realistic grid models.
- **Security Innovation:** Developing new frameworks, such as quantum game theory, for proactive grid defence.
- **Hardware Development:** Advancing toward fault-tolerant quantum processors capable of running complex grid simulations.
- **Standardization & Collaboration:** Establishing quantum-safe grid communication protocols through academic-industry partnerships and international cooperation.

Conclusion & Strategic Recommendations

Key Takeaways:

- Quantum technologies offer **transformative potential** for smart grids
- Current research demonstrates **practical benefits** across all grid domains
- Early adoption provides **competitive advantage** and future-proofing

Immediate Action

Partnerships

Collaborate with quantum technology providers.



Workforce

Develop quantum skills through training programs.



Piloting

Implement quantum technologies in targeted applications.



Assessment

Evaluate current utility systems for quantum readiness.

